Developing the Optimal Method to Detect Auditory Hazards for the Visually Impaired

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Introduction & Literature Review

With over 43 million blind individuals in the world and over 1 million in the United States alone, visual impairment is a prevalent issue in our society (Nature, 2024). Blind individuals rely frequently on their sense of hearing to compensate for their absence of visual input. Research from the Journal of Neuroscience utilizing functional MRI (fMRI) to identify differences in the auditory cortex—the part of the temporal lobe responsible for processing auditory information—between sighted and non-sighted individuals when provided auditory stimuli yields significant correlation between visual impairment and an enhanced auditory sense (Huber, 2019). Furthermore, Rafael Kons, a researcher from the Department of Physical Education at the Federal University of Bahia, Brazil, explored how para-judo athletes with different origins of visual impairment adapt to their surroundings, particularly focusing on how auditory and tactile stimuli enhance their performance. The study compared athletes with congenital and acquired visual impairments, highlighting the adaptive strategies these individuals develop to navigate their environment. Notably, the research concluded that both groups — those with congenital and those with acquired visual impairments — performed significantly better in tasks that involved auditory cues compared to standard testing conditions, underscoring the crucial role that auditory stimuli play in helping the visually impaired compensate for their lack of visual input (Kons, 2022). The study spotlights how visually impaired individuals, regardless of whether their impairment was present from birth or developed later on, have learned to rely on alternative sensory cues — notably auditory signals — to enhance their spatial awareness and performance. While this research is tailored to the unique context of para-judo athletes, it offers valuable insights into how visually impaired individuals — whether in sports or daily life — adapt to rely on their enhanced hearing to navigate the world around them.

This adaptation, however, is not without its limitations. Although their heightened auditory sensitivity helps them stay aware of their environment, a blind individual's reliance does not always result in accurate perception. Giorgia Bertonati, a Ph.D. student at the Istituto Italiano di Tecnologia, concludes that "the absence of visual experience in early life increases the auditory system's preference for the time domain and, consequentially, affects the perception of speed through audition", indicating that individuals who have been visually impaired their whole life may misidentify certain auditory stimuli because of their lack of real-world experience connecting vision and audio inputs (Bertonati, 2021). Without being able to draw on visual references, a blind person processes sounds in an atypical manner that leads to potential misidentifications or altered perceptions of auditory stimuli. Supporting this concept, Andrew Kolarik, a researcher from the Vision and Eye Research Institute at the Anglia Ruskin University, in Cambridge, United Kingdom, explores the correlation between visual impairment and changes in auditory abilities, particularly focusing on how the severity of visual impairment affects spatial auditory and distance perception. His study concludes that the accuracy of auditory spatial judgments is directly influenced by the degree of visual impairment, with individuals experiencing more significant vision loss facing greater challenges in perceiving spatial and distance cues through hearing alone (Kolarik, 2020).

This issue becomes particularly pressing in public environments that are noisy and chaotic. The presence of loud noises such as traffic noise, construction, and conversation among others may drown out important auditory cues that would otherwise indicate potential hazards to visually impaired individuals (Iberdrola, 2024). For those who have lost vision early in life, their ability to integrate auditory signals with other sensory inputs is further impaired in public environments. University of Oxford researcher Andrew King's work notably concluded that "early loss of vision has been reported to impair the ability of the central nervous system to combine and integrate multisensory cues" and "they [visually impaired individuals] usually have difficulty in localizing sounds and perceiving speech in noisy conditions" (King, 2008). Without the visual cues that many people rely on to contextualize and differentiate sounds, these individuals may find it difficult to discern the direction or distance of important—and potentially hazardous—auditory signals. The difficulty in processing such auditory cues in noisy environments may lead to confusion, disorientation, and potential danger for the individual. As a result, those who lost their vision early in life are especially vulnerable to environmental hazards that rely on the integration of multiple sensory inputs for accurate detection, making it even harder for them to safely navigate noisy public spaces.

Current devices to assist the visually impaired in such noisy environments do not adequately resolve the underlying problem of ensuring their safety while navigating in public. Harry Levitt, PhD at The Lexington School for the Deaf, offers a comprehensive review of current methods to reduce background noise in traditional hearing aids, focusing particularly on techniques aimed at enhancing speech recognition in noisy environments. Levitt explains how audio processing within hearing aids maintains a strong focus on the different types of background noise — such as environmental hums, chatter, and other distracting sounds — and how they are filtered out to improve the clarity of speech (Levitt, 2001). Noise reduction algorithms and directional microphones are utilized to prioritize sounds that are most relevant to the user, making it easier for individuals to focus on speech and reduce the impact of competing noise. Unfortunately, such devices do not serve much benefit for the visually impaired, as their use case is focused on person-to-person conversation, as opposed to amplifying important auditory cues to enhance the safety of the user. The long-term use-case for hearing aids is explored by researcher Mahsa Pouyandeh from the Department of Audiology at the Tehran University of Medical Sciences, whose work emphasizes that the primary satisfaction with hearing aids is strongly correlated with the length of time a user has been wearing the device (Pouyandeh, 2019). Over time, as individuals become more accustomed to the hearing aid, they generally report higher levels of satisfaction with its performance and utility. This insight raises important consideration for visually impaired individuals who may only use hearing aids in specific circumstances, such as in noisy public environments. In such cases, where the hearing aid is only needed temporarily, these users may not experience the same level of satisfaction as long-term users. This discomfort is crucial to take account of, especially when incorporating a piece of technology into the daily lives of such individuals.

Juan Gomez, Computer Science Phd. (computer vision & neuroscience) from the University of Geneva, presents a novel sensory substitution device designed for the visually impaired which utilizes a camera to detect, classify, and communicate objects to the user through bone-conduction headphones, effectively providing a way for visually impaired individuals to perceive their environment (Gomez, 2014). The technology relies primarily on computer vision techniques to process visual data, converting it into sensory information that can be transmitted through auditory channels. Importantly, however, the device does not utilize auditory data or enhance the user's awareness through auditory input alone; it instead relies entirely on vision-based object detection, which completely disregards any potential auditory cues that the user may not pick up naturally.

Despite these significant advancements in assistive technology for the visually impaired, a clear gap remains in addressing the specific needs of individuals navigating noisy public environments. Existing technologies, such as hearing aids and visual-object recognizers, do not adequately address the masking of potential auditory cues for hazards being drowned out by noisy public environments. There is a pressing need for a novel device specifically designed to detect and communicate auditory hazards in noisy public environments.

This urges the question, what is the most optimal way to detect auditory hazards in noisy conditions and communicate them with a visually impaired user to improve their awareness and safety in noisy environments?

Methodology

Machine learning is the subfield of artificial intelligence in which a machine makes predictions from data. Depending on the type of data provided and algorithms used, machine learning models will vary in accuracy and efficiency when applied to a classification task. In the context of audio classification, machine learning models are trained on audio data and then tasked with identifying the category or class of foreign audio samples. This process involves feature extraction, where relevant characteristics of the audio signal are converted into a form suitable for feeding into machine learning algorithms.

The core of the methods used in this project involves Librosa, a widely-used audio analysis library that extracts unique audio features, using them as fingerprints to identify each sound sample. Librosa generates feature vectors from raw audio signals, transforming each audio clip into a unique, high-dimensional representation. These features include spectral representations such as the zero-crossing rate, centroid, bandwidth, contrast, rolloff, root-mean-square, chroma, and Mel-frequency cepstral coefficients that describe the tonal and temporal properties of the sounds, giving them unique “fingerprint” vectors. These identities are then used as inputs to a neural network, or machine learning model, which learns to map these representations to the corresponding sound class labels.

This method aligns with the current body of research where machine learning is applied to audio data. Research has shown that neural network architectures originally designed for image classification, like VGG, Inception, and ResNet, can be effectively adapted for audio classification tasks. These models have demonstrated impressive results when trained on large-scale datasets, with some studies utilizing up to 5.24 million hours of training videos across 30,871 labels (Hershey et al., 2017). Neural networks are particularly well-suited for this task because of their ability to learn hierarchical patterns in data, which, in the case of audio, can include both low-level spectral features and high-level patterns that are crucial for classification. Additionally, using Librosa to extract robust features provides a more flexible and efficient approach compared to raw waveform-based methods, allowing for better performance in many real-world classification tasks. By leveraging techniques such as Mel-frequency cepstral coefficients (MFCCs), chroma features, spectral contrast, and zero-crossing rate, Librosa enables the transformation of complex audio signals into meaningful representations that highlight relevant patterns for machine learning models. This leads to such models developing the capacity to capture essential characteristics of the audio, such as pitch, timbre, rhythm, and energy, to allow more stark contrast between different classes, thereby easing classification. This approach not only reduces the dimensionality of the input data but also minimizes the impact of noise and variability in the audio signals.

Evaluating the performance of a sequential model for audio classification is crucial to understanding its effectiveness in accurately classifying amongst the seven distinct classes. While accuracy is a fundamental metric, it does not always provide a comprehensive picture. Therefore, precision, recall, F1-score, and confusion matrices were used cooperatively to get a more well rounded understanding of each model.

Precision measures the proportion of correctly predicted instances of a given class out of all instances predicted as that class. This metric is particularly useful when false positives need to be minimized. In this particular research, high precision indicates that a model's classification is highly likely to be correct.

Recall, on the other hand, quantifies the proportion of correctly identified instances of a class out of all actual instances of that class. This metric is crucial when false negatives need to be minimized. In the context of audio classification, a low recall value would indicate that the model is failing to recognize many instances of a particular class, reducing its reliability in real-world scenarios.

The F1-score, which is the harmonic mean of precision and recall, serves as a balanced metric that provides insight into both false positives and false negatives. By incorporating both metrics, the F1-score ensures that the model's performance is not overly biased toward precision or recall, making it a robust indicator of classification effectiveness across all classes.

Additionally, confusion matrices provide a visual representation of the model's predictions compared to the actual labels. Each row of the matrix represents the true classes and each column represents the predicted classes. By analyzing the confusion matrix, misclassifications and patterns of errors can be easily discerned. When combined, these evaluation metrics ensure that each model is assessed comprehensively.

The UrbanSound8k dataset from Kaggle was used for training and testing the model. This dataset contains 8732 labeled audio clips from 10 different urban sound classes. For the purposes of this project, the dataset was narrowed to focus on 7 specific sound classes: Car Honk, Dog Barking, Drilling, Engine Idling, Jackhammer, Siren, and Violence. From the UrbanSound8k dataset, these classes were the most pertinent to a real-world scenario involving the safety of a visually impaired pedestrian. Each audio sample is in a .wav format, providing uncompressed, high-quality audio data. The files are under 4 seconds and are distributed evenly across the 7 selected classes. The dataset contains around 870 instances per class, providing sufficient data for training, validation, and testing purposes.

The features extracted from the audio files were designed to capture the most relevant characteristics of the sounds, which are critical for classification. These features were obtained using Librosa and include a combination of spectral and temporal features that describe the frequency content, rhythm, and pitch of the sound. Key features included spectral centroid, which represents the "center of mass" of the spectrum and is useful for identifying the timbral quality of sounds; spectral contrast, which measures the difference in amplitude between spectral peaks and valleys, reflecting the texture of a sound; and chroma features, which are essential for capturing harmonic and tonal content.

Additionally, Mel-frequency cepstral coefficients (MFCCs) were computed to represent the perceptual characteristics of the sound, similar to how human hearing perceives sound. Zero-crossing rate, another temporal feature, was also included, which indicates how often the signal changes sign, capturing noisiness and transient events in the sound. These features are thorough representations of the underlying audio data and are essential for distinguishing between different classes of sounds in urban environments.

In total, five different sequential deep learning models were developed and tested to determine the optimal model architecture. All models were implemented using TensorFlow and Keras in Python.

The first model serves as a baseline, utilizing fully connected dense layers for classification, establishing a benchmark for classification accuracy. The simplicity of this model causes it to lack feature extraction capabilities, leading to potential overfitting on the dataset. This model's architecture can be seen in Figure 1.

Fig. 1 ***Baseline DNN Architecture***

```
Dense(512,
activation='relu', ),
   Dense(512/2,
activation='relu'),
   Dense(512,
activation='relu'),
   Dense(512/2,
activation='relu'),
   Dense(512,
activation='relu'),
   Dropout(0.1),
   Dense(7,
activation='softmax')
```

The second model incorporates Batch Normalization — a technique used to stabilize learning and prevent overfitting — which improves loss convergence speed and improves generalization over the dataset. This model's architecture can be seen in Figure 2.

Fig. 2 ***Batch Normalized DNN Architecture***

```
Dense(512,
activation='relu'),
   BatchNormalization(),
   Dense(256,
activation='relu'),
   BatchNormalization(),
   Dense(128,
activation='relu'),
   Dropout(0.2),
   Dense(7,
activation='softmax')
```

The third model incorporates Long Short-Term Memory (LSTM) layers to capture the sequential patterns and information present in the audio signals. LSTMs are designed to handle sequential data (gene sequences, video clips, audio signals) and are unique as they maintain memory of previous time steps. This ability to remember and process time-based information is especially useful for audio classification, where the order and progression of sounds matter. This model can detect patterns present in the gradual build-up of noise and rising pitch. This model's architecture can be seen in Figure 3.

Fig. 3 ***LSTM Model Architecture***

```
LSTM(128,
return_sequences=True),
   LSTM(64),
   Dense(128,
activation='relu'),
   Dropout(0.3),
   Dense(7,
activation='softmax')
```

The fourth model combines Convolutional Neural Networks (CNNs) for feature extraction and LSTMs for sequential analysis. The convolutional layer (Conv1D), which applies filters to the input feature maps, allows the network to learn local patterns. The following max-pooling layer (MaxPooling1D) is applied to reduce the spatial dimensions of the feature maps, which helps prevent overfitting and improves the model's computational efficiency. This model's architecture can be seen in Figure 4.

Fig 4. ***Hybrid CNN-LSTM Model Architecture***

```
Conv1D(64,
kernel_size=3,
activation='relu'),

MaxPooling1D(pool_size=
2),
   LSTM(64,
return_sequences=True),
   LSTM(32),
   Dense(128,
activation='relu'),
   Dropout(0.2),
   Dense(7,
activation='softmax')
```

The fifth model increases the depth of the dense layers in the first model, providing more abstraction but with a lower learning rate to prevent loss divergence. The purpose of this model is to explore whether a deeper network can improve classification accuracy. This model's architecture can be seen in Figure 5.

Fig. 5 ***Deeper DNN Architecture***

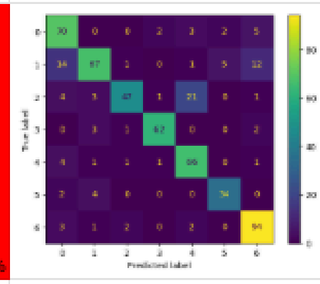
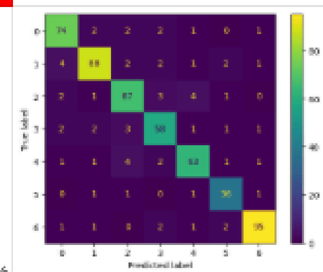
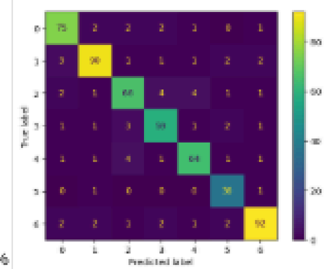
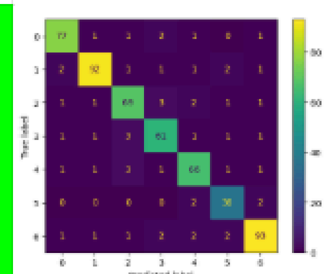
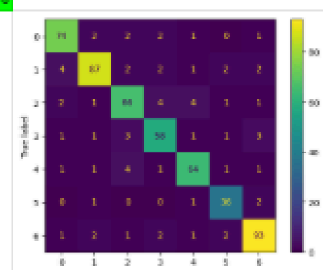
```
Dense(1024,
activation='relu'),
   Dense(512,
activation='relu'),
   Dense(256,
activation='relu'),
   Dense(128,
activation='relu'),
   Dropout(0.3),
   Dense(7,
activation='softmax')
```

This project was conducted with the assistance of a few core libraries and software tools. The underlying programming language used was Python. This was chosen because it is widely favored in the machine learning community for its extensive support for libraries and ease of use for programmers. Jupyter Notebook was the environment utilized, as it allows for interactive coding and ease of experimentation. Jupyter Notebook notably allows for efficient visualization of dataframes. The Pandas library was used for data manipulation, particularly organizing extracted audio features into structured data-frames that could readily be processed and fed into the machine learning models.

It is also important to note that the different model types used in this research require different forms of data in order to function properly. The Dense architectures require the input data to be transformed into flattened, fixed-size, 1D vectors. This means each audio clip needs to be compressed into a single, consistent feature vector. For this task, Mel-Frequency Cepstral Coefficients (MFCCs) were averaged across each audio clip to create a 1D vector. LSTM models, on the other hand, require sequential data, as they process inputs over time. Instead of using a single average vector, one average MFCC vector was calculated every 10 milliseconds, maintaining the temporal aspect of the audio data. Convolutional Neural Networks require 2D inputs. Therefore, a spectrogram matrix was utilized, representing how the audio signals' frequencies change over time in a 2-dimensional manner. All of the data manipulation, including extraction of the MFCCs and conversion to spectrograms, was done using the Librosa Python package.

The deep learning models presented in this paper were developed using TensorFlow. This open-source library offers user-friendly functions and models to implement machine learning algorithms with ease. TensorFlow provided the necessary tools to define and train the different sequential layers, such as the Conv1D for learning spatial patterns and the MaxPooling1D for dimensionality reduction. Dropout was also implemented for the purpose of regularization in order to prevent overfitting and enhance generalization.

Fig. 6 ***Confusion Matrices, Accuracy, Precision, Recall, and F1-Scores for Each Model***

Model Architecture	Optimizer	Loss	Accuracy Score	Precision	Recall	F1-Score	Confusion Matrix
Dense(512, activation='relu',), Dense(512/2, activation='relu'), Dense(512, activation='relu'), Dense(512/2, activation='relu'), Dense(512, activation='relu'), Dropout(0.1), Dense(7, activation='softmax')	Adam	Categorical Crossentropy	81.03%	82.42%	81.56%	81.16%	
Dense(512, activation='relu'), BatchNormalization(), Dense(256, activation='relu'), BatchNormalization(), Dense(128, activation='relu'), Dropout(0.2), Dense(7, activation='softmax')	Adam	Categorical Crossentropy	88.58%	87.84%	88.41%	88.10%	
LSTM(128, return_sequences=True), LSTM(64), Dense(128, activation='relu'), Dropout(0.3), Dense(7, activation='softmax')	Adam	Categorical Crossentropy	88.77%	87.99%	89.04%	88.44%	
Conv1D(64, kernel_size=3, activation='relu'), MaxPooling1D(pool_size=2), LSTM(64, return_sequences=True), LSTM(32), Dense(128, activation='relu'), Dropout(0.2), Dense(7, activation='softmax')	Adam	Categorical Crossentropy	90.48%	89.71%	90.29%	89.97%	
Dense(1024, activation='relu'), Dense(512, activation='relu'), Dense(256, activation='relu'), Dense(128, activation='relu'), Dropout(0.3), Dense(7, activation='softmax')	Adam	Categorical Crossentropy	87.87%	87.30%	87.85%	87.55%	

Data Analysis

The following accuracy metrics mentioned can be viewed in Figure 6.

The baseline dense neural network performed with a classification accuracy of 81.03%. This model's low precision and recall scores indicate that it struggled to generalize across all classes, producing a lower F1-score as a result. As the basis of this model is a simple feed-forward network, these accuracy metrics suggest that such an architecture may not be well suited for capturing the sequential patterns that exist in audio data. The performance of this model highlights a clear need for the model to gain the capacity to learn temporal dependencies, which are a necessity in time-series classification for tasks like audio recognition.

Introducing batch normalization proved fruitful, as the accuracy score for this model jumped to 88.58%. As batch normalization works to stabilize training, this procedure likely helped the model achieve better overall generalization. The precision and recall scores indicate that the model is able to classify instances more reliably than the baseline model, which is also backed by a heightened F1-score when compared to the baseline model. These improvements suggest that, although the model is still fundamentally a feed-forward network, batch normalization can help mitigate some of the issues associated with training simple neural networks and can improve performance for classification tasks with time-dependent data.

The LSTM model achieved an accuracy score of 88.77%. It can be reasonably inferred, by the nature of LSTM architectures, that this model was able to capture long-term dependencies within the time-series audio data, proving its dominance over the previous dense neural networks. The recall and F1-scores demonstrate that the LSTM is able to grasp a more comprehensive understanding of the audio signals because of its ability to better model the sequential nature of the audio signals. This improvement is a vital leap to further solidifying the optimal model.

The hybrid CNN-LSTM model performed with not only the highest accuracy score of 90.48%., but also the highest precision, recall, and F1-scores. It is evident that combining the strengths of the convolutional neural networks for spatial feature extraction and LSTMs for sequential processing allow this model to perform classification on audio signals the most effectively. The success of this hybrid model highlights the importance of using both feature extraction and sequential modeling simultaneously to handle the complexities of time-series data, particularly audio data in this case.

Increasing the depth of the original baseline deep neural network by incorporating additional layers with 1024 and 512 units yielded an accuracy score of 87.87%, which was slightly lower than the LSTM and CNN-LSTM architectures. This score provides valuable insight into the impact of network depth on classification performance. The precision and recall scores suggest that while the model performs well, it does not benefit greatly from the significant increase in depth. A possible explanation for this tapering accuracy is overfitting, which occurs when the model trains too well on the provided training data, and in turn performs worse on foreign data. The deeper network may have struggled to generalize as effectively as the simpler or hybrid models because the added complexity likely introduced challenges in training stability\. This suggests that in contexts regarding sequential data like audio files, a deeper neural network may not necessarily lead to better performance. Other factors, like feature extraction and temporal sequencing, will provide more value for tasks like audio classification.

Limitations

The methodology and research design process in this paper comes with limitations to consider. Although the classes chosen from the UrbanSound8k dataset contain balanced samples for each, the actual audio files for some classes may have been more complex than others. Subtle variations in this complexity may lead to model bias when predicting. An example of this is the way a particular sound exists in real life (e.g. a siren having an easily discernible high pitch). An additional point of constraint for this methodology is the feature extraction procedure process itself. The MFCCs and spectral centroid features are effective and vital in capturing the key distinguishing aspects of sound, especially in the context of being fed into a machine learning algorithm. Unfortunately, these features are not able to fully capture the nuances of raw audio signals from the real-world. After all, an analog representation of real-world sounds is bound to leave out information. Computational complexity is an additional concern, as training neural networks on large datasets comes with the expense of GPU power, which can be expensive when model complexity and dataset size are increased significantly.

Conclusion

Overall, the hybrid CNN-LSTM model took the crown as the best-performing architecture, achieving the highest accuracy and delivering strong generalization across all classes. This supports the idea that integrating spatial and temporal modeling strategies—through LSTM—alongside the feature extraction capabilities—through CNN—can lead to more effective learning, especially on complex sequential data like audio files. It is also important to note that the feature extraction aspect of this CNN-LSTM hybrid model helps to reduce the dimensionality of the input, further focusing the model's attention on the most informative aspects of the audio data, a likely component of the high classification accuracy. On the other hand, we saw that dense networks, despite being augmented to incorporate deeper layers and batch normalization, consistently struggled with sequential patterns. This limitation made them less suitable for this classification task, proving that model depth alone didn't offer a significant boost to classification accuracy.

Moving forward, there are lots of ways to further improve upon the accuracy and efficiency of the hybrid CNN-LSTM model. Future work could explore more nuanced hyperparameter tuning to optimize each of the CNN and LSTM layers. Hyperparameter tuning could be performed by a program curated to test different parameters for a baseline model, and evaluating which combination of hyperparameters produces the highest accuracy. Experimenting with different architectures within the domain of this hybrid model, such as adding attention mechanisms or exploring normalization techniques, could help the model hone in on more relevant features of the audio signals. Additionally, data augmentation could be applied to increase the size of the provided dataset, making the model more robust to variations in audio files. This augmentation could include the addition of background noise or continuous recording changes, making the dataset more congruent to real-world scenarios.

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